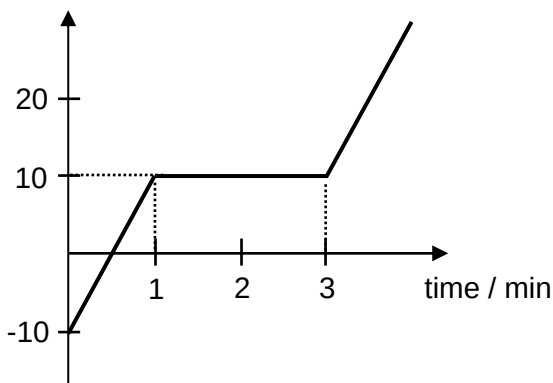


- 12** A student heats a 500 g solid sample at an initial temperature of -10°C . The rate of heat absorbed by the sample is constant at 200 W. The graph below shows how the temperature of the sample varies with time.

temperature / $^{\circ}\text{C}$



What is the specific latent heat of fusion of the solid sample?

- | | | | | | | | |
|----------|------------------------|----------|------------------------|----------|------------------------|----------|------------------------|
| A | 12 kJ kg^{-1} | B | 18 kJ kg^{-1} | C | 36 kJ kg^{-1} | D | 48 kJ kg^{-1} |
|----------|------------------------|----------|------------------------|----------|------------------------|----------|------------------------|

Answer: D

Energy supplied $E = Pt$ where P is the power supplied in time t .

By conservation of energy, $E = Pt = mL$ where m is the mass and L the specific latent heat of solid.

$$200(2 \times 60) = 0.500L$$

$$L = 48000 \text{ J kg}^{-1}$$

$$= 48 \text{ kJ kg}^{-1}$$

- 13** The density of argon gas at a pressure of $1.00 \times 10^5 \text{ Pa}$ is 1.60 kg m^{-3}